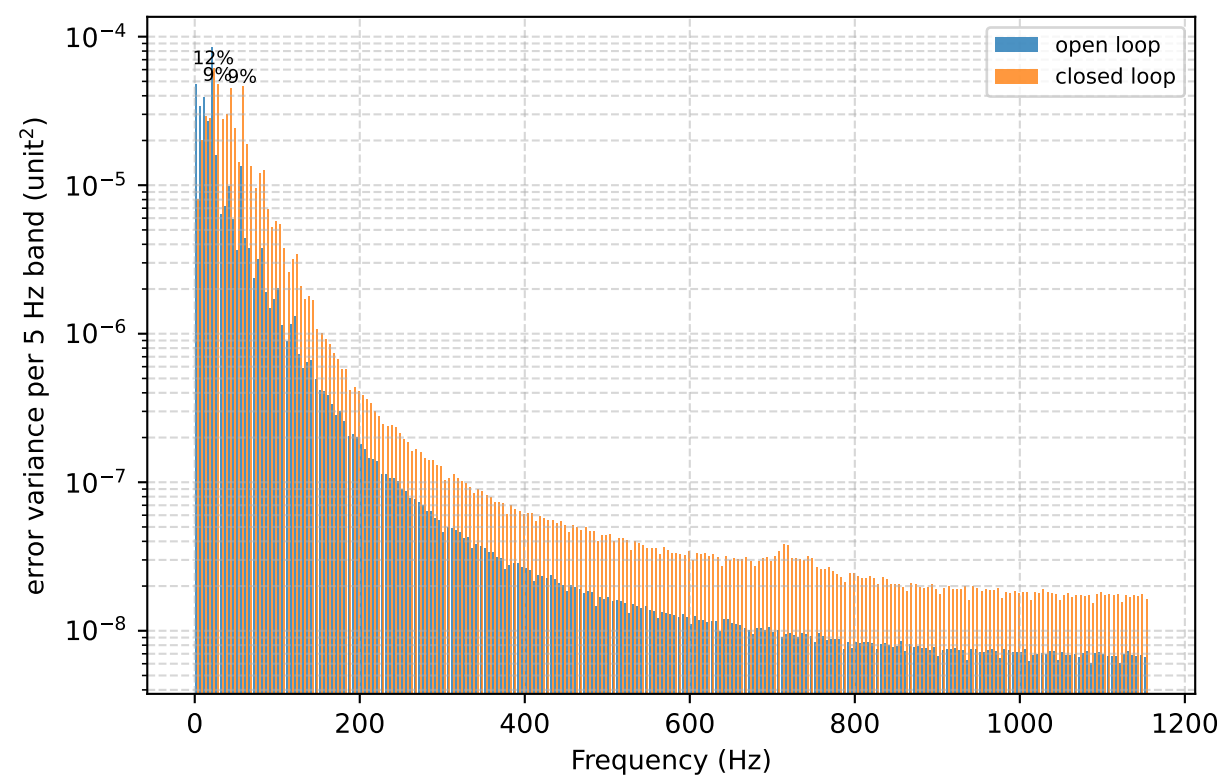
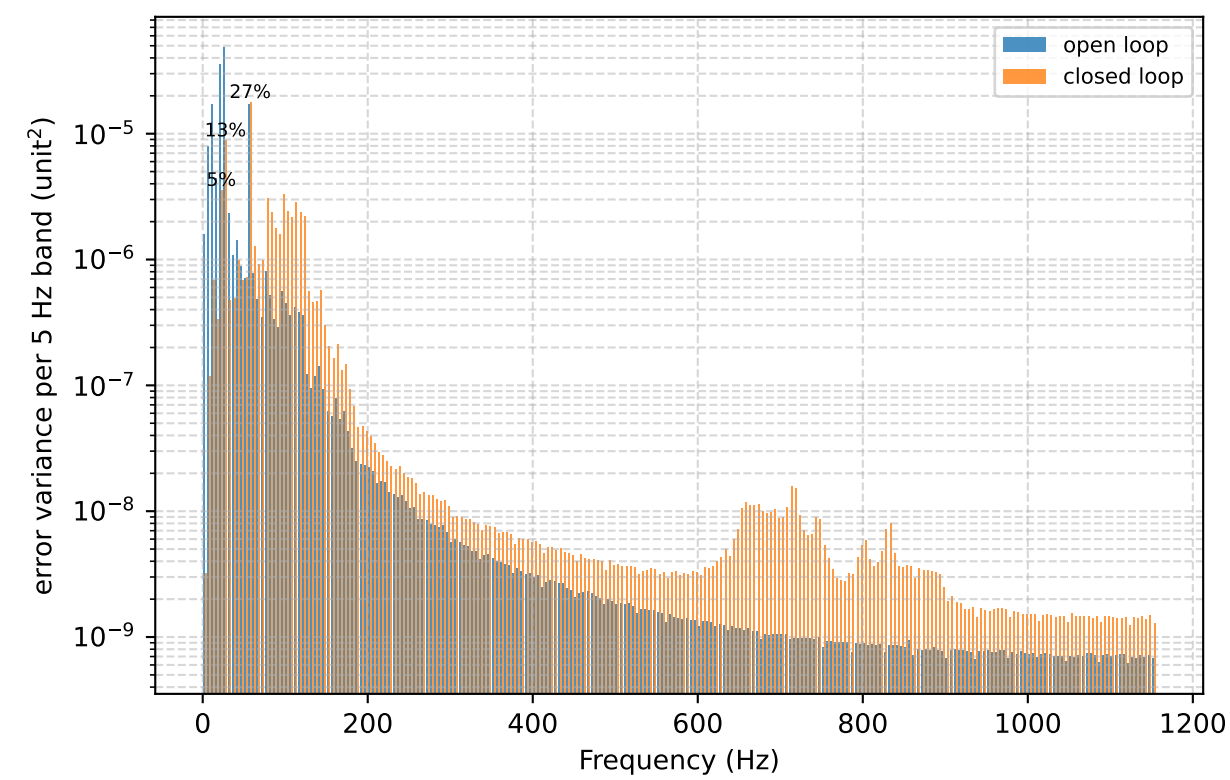
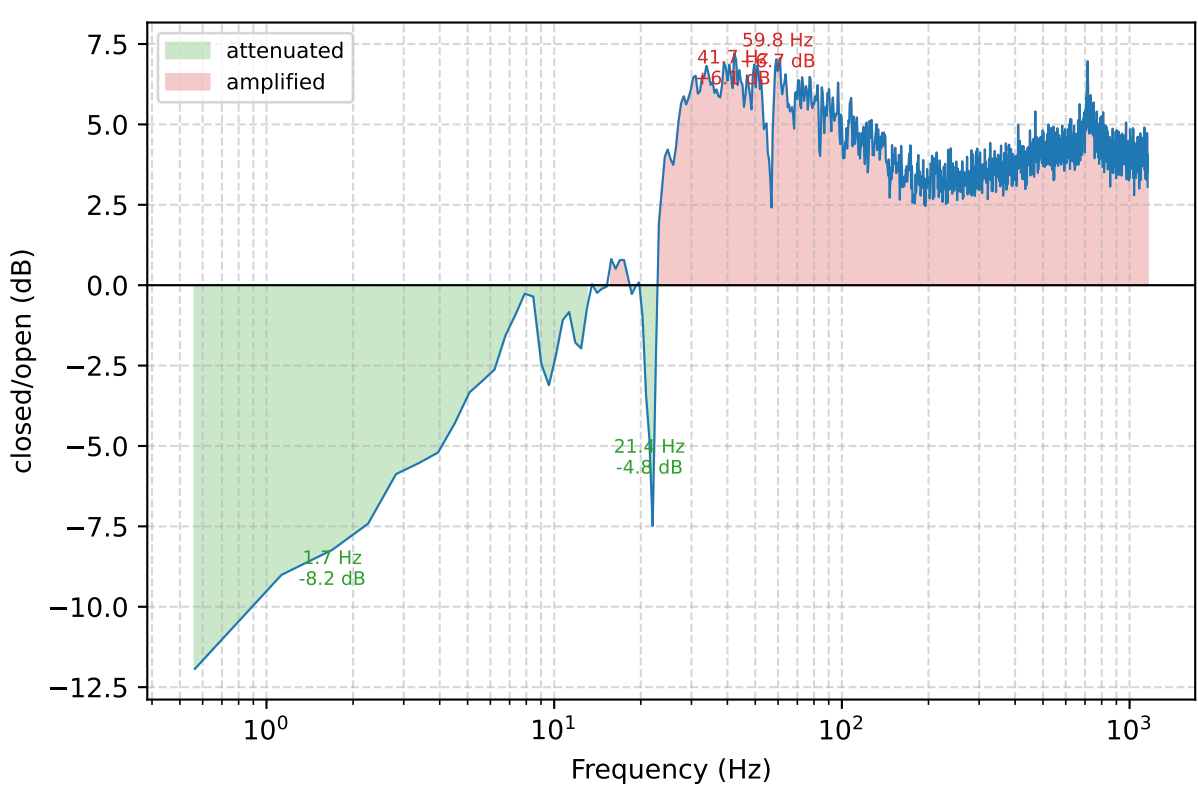
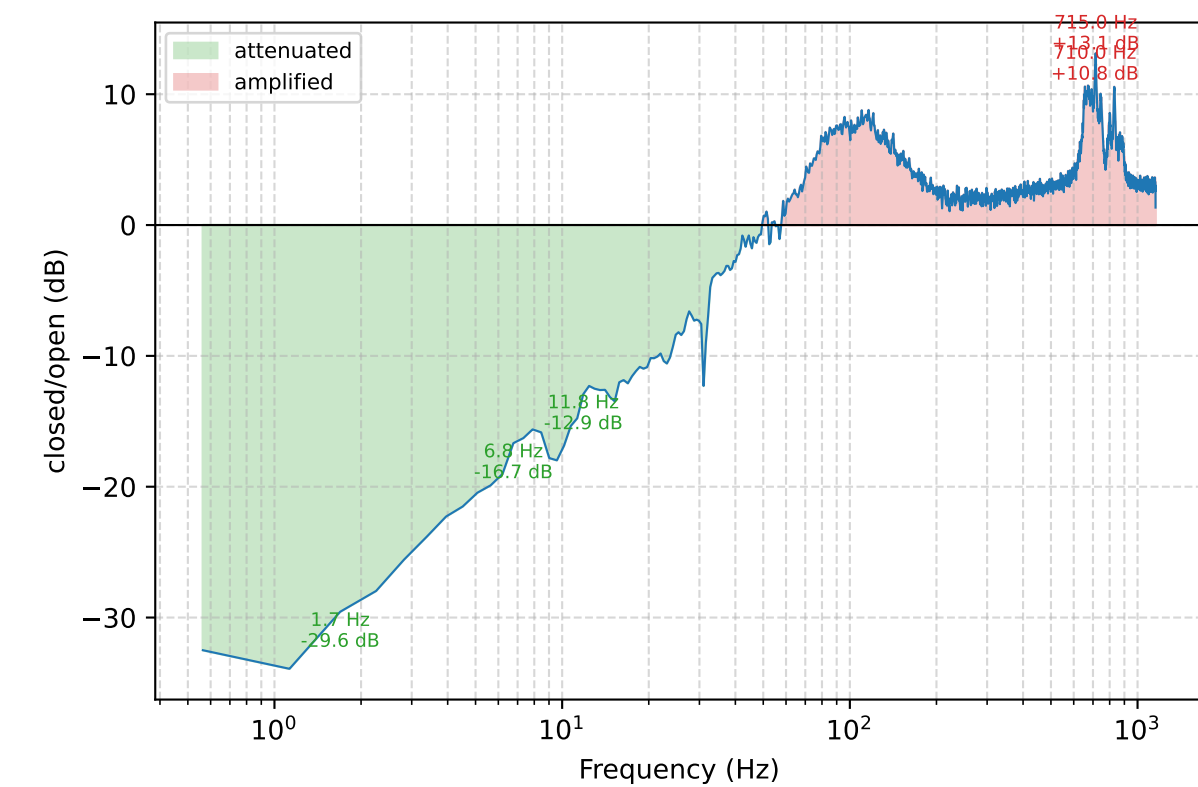
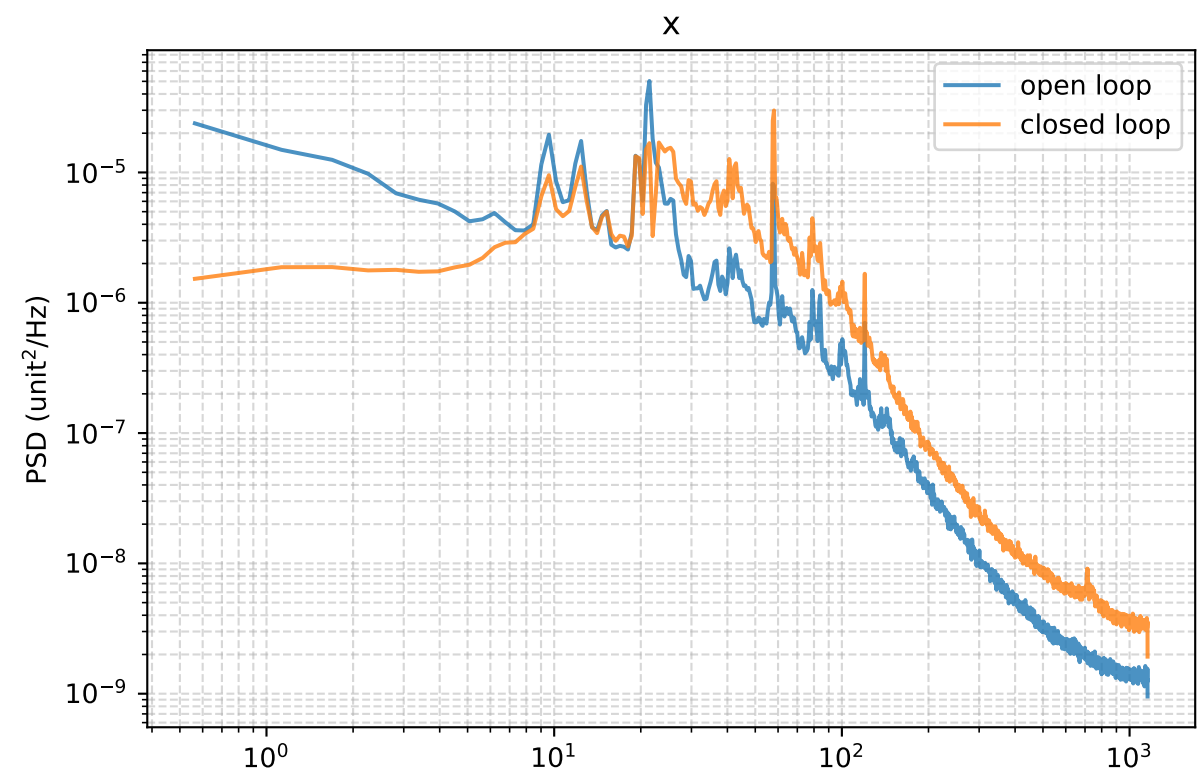
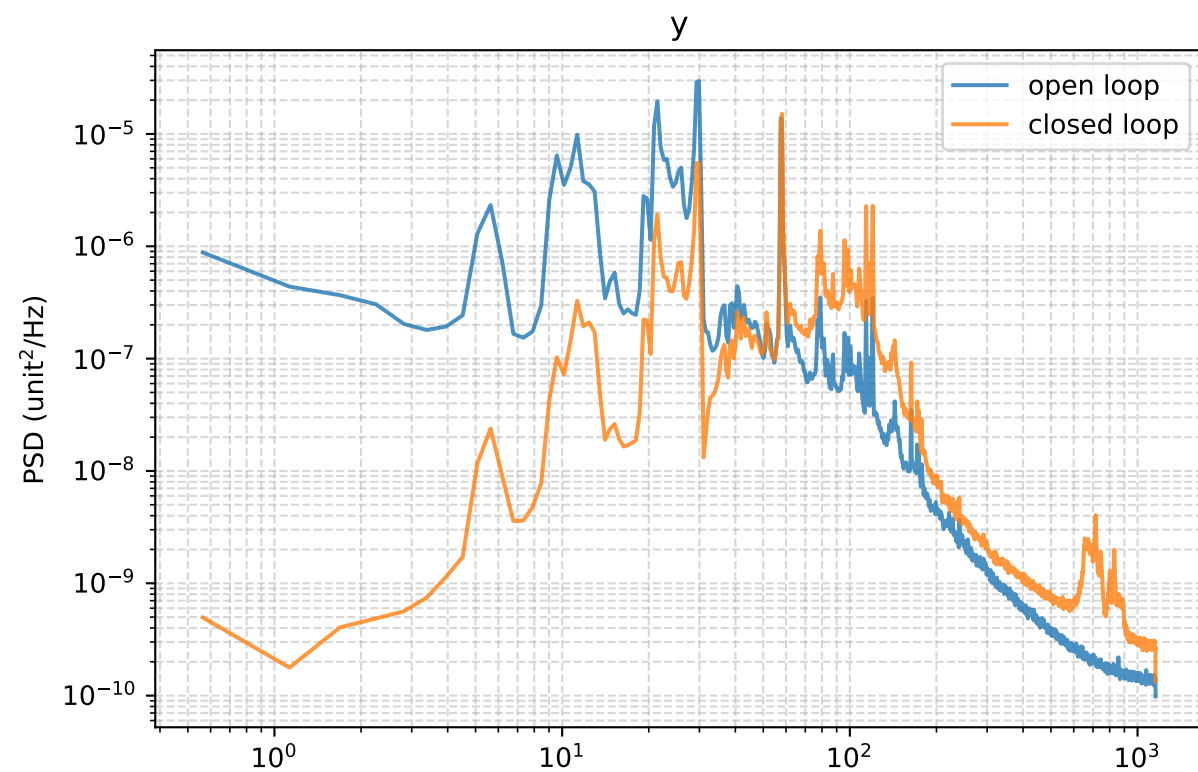


Closed-loop attenuation



Run configuration and per-band RMS

Started: 2026-07-13 14:12:00
pixel_scale: 31.5 px per output unit
y: mean +2.221 -> -0.000 px; RMS about mean 0.435 -> 0.257 px (open -> closed)
x: mean -1.782 -> -0.004 px; RMS about mean 0.761 -> 0.713 px (open -> closed)

Config used:
100, 500, 100, 500, 4096: camera ASI290MM 64x64, ROI start (920,488), ~2310 Hz loop
pid: integrator-only (p=py=d=0), i=600 iy=1300, units=minmax
lines x=[22.08,57.3] gain=[60,20] phase=[56,3]
linesy=[30.8] gainy=[60] phasey=[60], line_delay=0.002
kalman: order=60 horizon=6 mu=0.03 gain=0.7 gainy=0 ramp=10
piplate: fine ch0(+2)/ch1(-2), coarse parked, baud 460800, start_delay: 100.0, open_time: 500.0, settle_time: 100.0, closed_time: 500.0

RMS contributed by each 10 Hz band (px)				
band (Hz)	y open	y closed	x open	x closed
0-10	0.0975	0.0110	0.2845	0.1672
10-20	0.1465	0.0318	0.2559	0.2384
20-30	0.2885	0.1112	0.3153	0.3288
30-40	0.0583	0.0311	0.1157	0.2393
40-50	0.0479	0.0407	0.1248	0.2624
50-60	0.1327	0.1359	0.1302	0.2445
60-70	0.0353	0.0466	0.0900	0.1788
70-80	0.0339	0.0633	0.0740	0.1466
80-90	0.0291	0.0640	0.0750	0.1389
90-100	0.0290	0.0698	0.0564	0.1039
100-110	0.0283	0.0674	0.0560	0.0958
110-120	0.0282	0.0720	0.0451	0.0756
120-130	0.0218	0.0524	0.0449	0.0739
130-140	0.0146	0.0302	0.0350	0.0588
140-150	0.0153	0.0293	0.0338	0.0522
150-160	0.0108	0.0192	0.0286	0.0438
160-170	0.0115	0.0185	0.0266	0.0397
170-180	0.0102	0.0155	0.0240	0.0352
180-190	0.0075	0.0106	0.0214	0.0313
190-200	0.0068	0.0095	0.0202	0.0290
200-210	0.0065	0.0086	0.0186	0.0272
210-220	0.0058	0.0075	0.0169	0.0252
220-230	0.0055	0.0069	0.0158	0.0228
230-240	0.0051	0.0066	0.0148	0.0218
240-250	0.0050	0.0062	0.0143	0.0211
250-260	0.0046	0.0059	0.0133	0.0194
260-270	0.0041	0.0053	0.0124	0.0180
270-280	0.0040	0.0052	0.0119	0.0174
280-290	0.0039	0.0049	0.0112	0.0166
290-300	0.0038	0.0048	0.0106	0.0160
total 0-300	0.3805	0.2560	0.5733	0.7070